Lecture 2: Sequencing Technology for BigData in Cancer Biology

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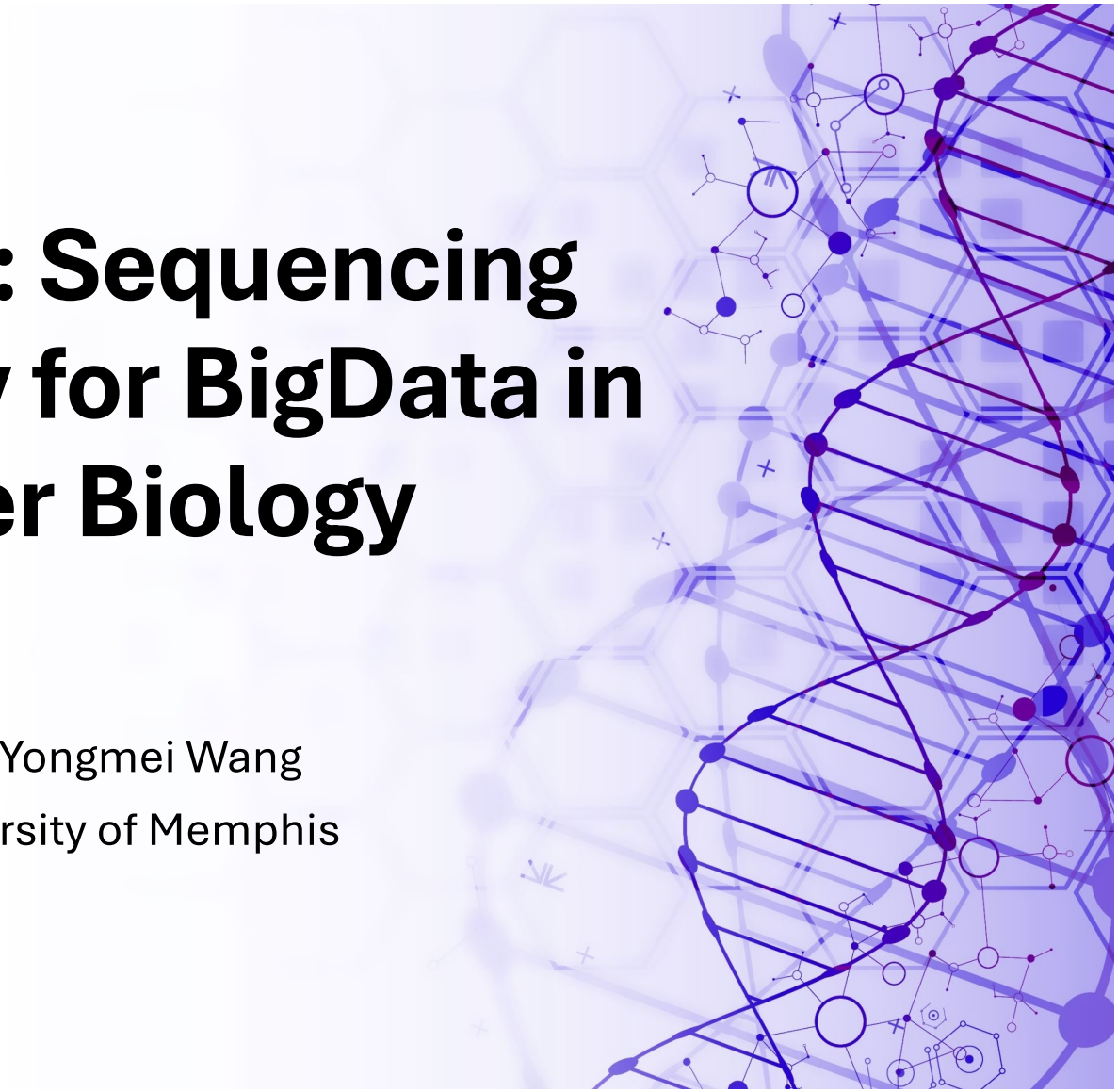


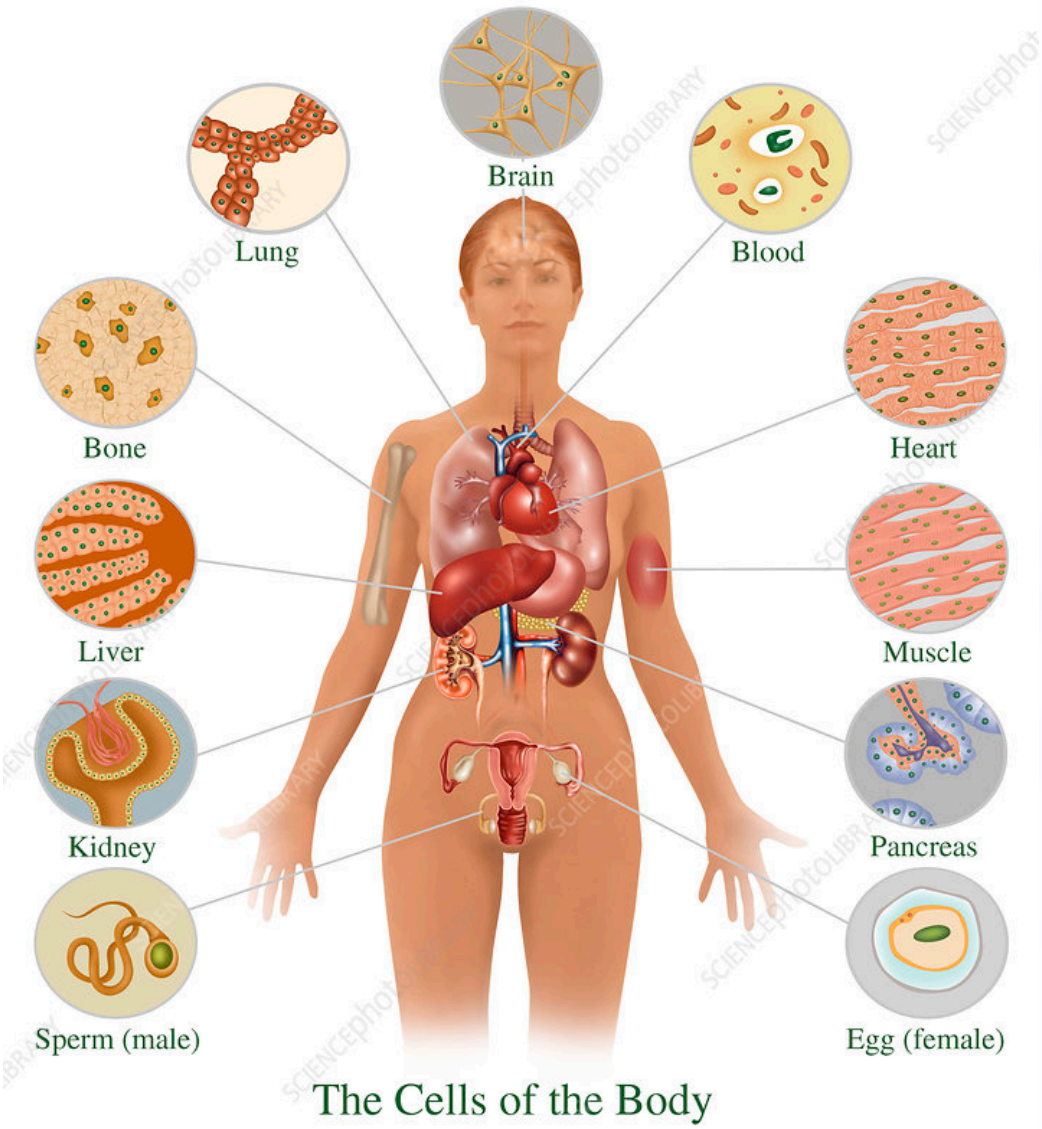
Activity: Jupyter Notebook

- Install and launch Jupyter notebook
- Use Jupyter notebook to plot some functions, y=x^2
- Use Jupyter notebook to plot some data contained in a file

Big data in cancer biology

- It happened after the human genome project completed (~2003).
- The Cancer Genome Atlas Program (TCGA) launched in 2006
- Big data fueled by the **low costs of sequencing**

https://www.cancer.gov/ccg/research/genome-sequencing/tcga



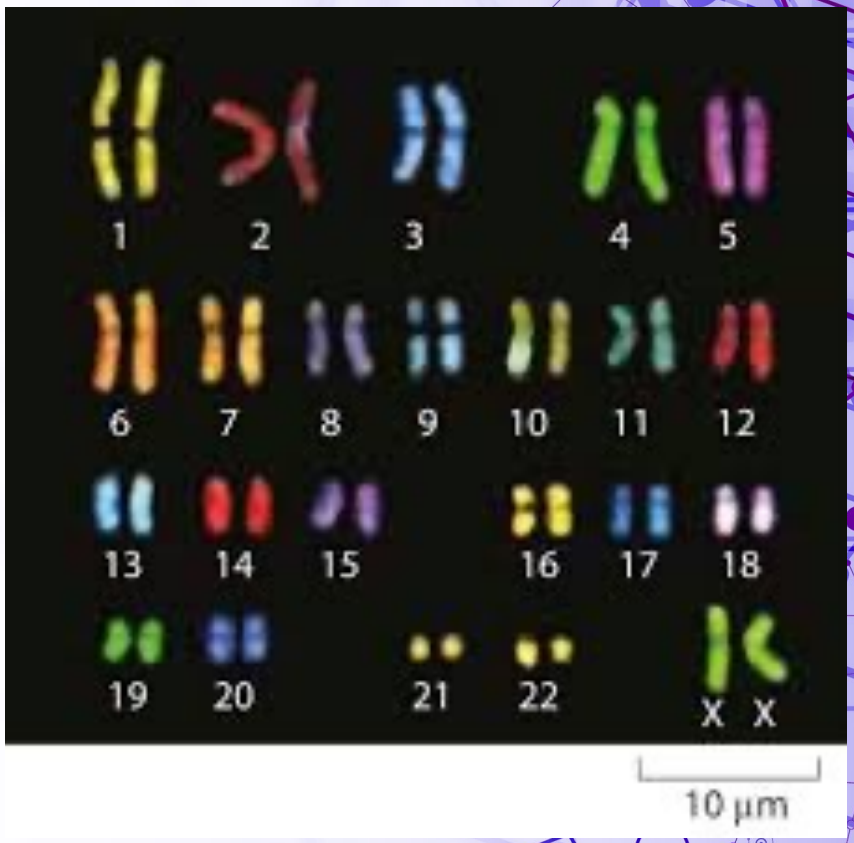
Human Cells

All these cells have approximately "same DNA genome"

https://www.sciencephoto.com/media/1136879/view/cells-of-the-body-illustration

Basic genetics of human

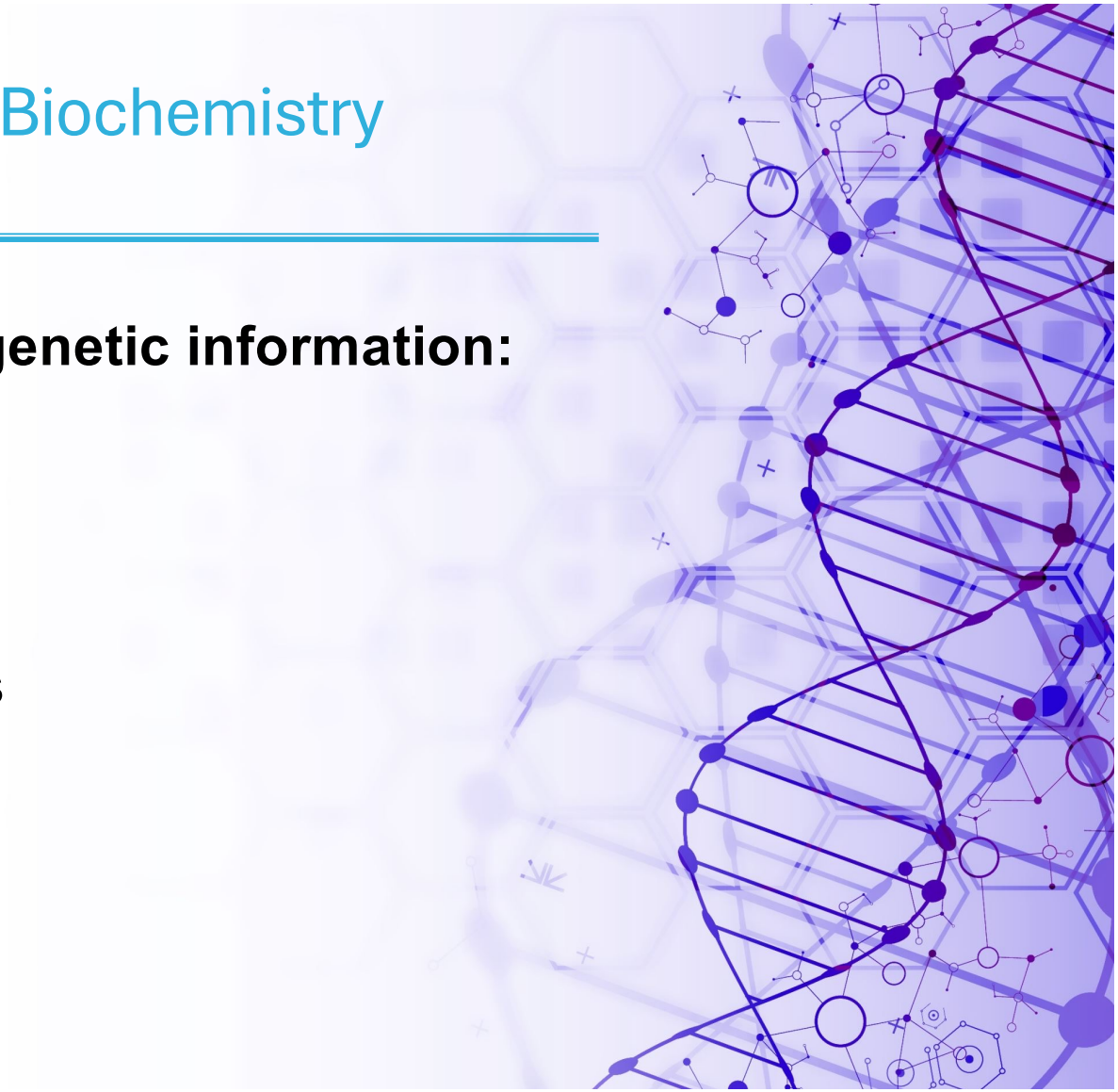
- Human has 23 pairs of chromosome
- Human genome project – sequences all DNA bp in human genome :3 billions base pairs (bps) ($3x10^9$ bps)
- Provide a human genome references

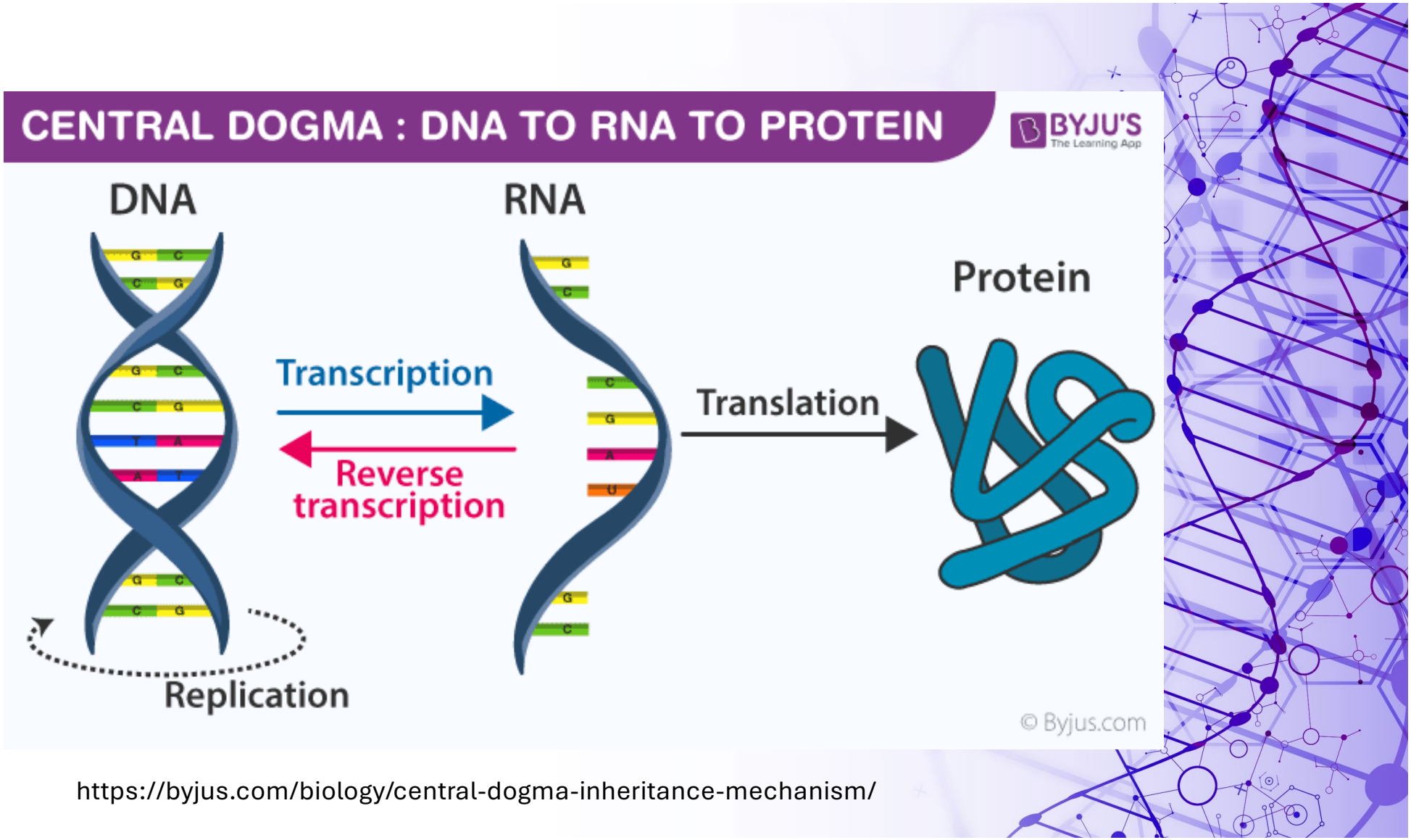


The Central "Dogma" of Biochemistry

Pathway for the flow of genetic information:
DNA → RNA → Protein

DNA stores information
RNA transmits information
Protein function manifests information





https://byjus.com/biology/central-dogma-inheritance-mechanism/

Discovered DNA double helix structure in 1953

James D. Watson

Francis Crick, 1916–2004

Unnumbered 8 p277
Lehninger Principles of Biochemistry, Fifth Edition
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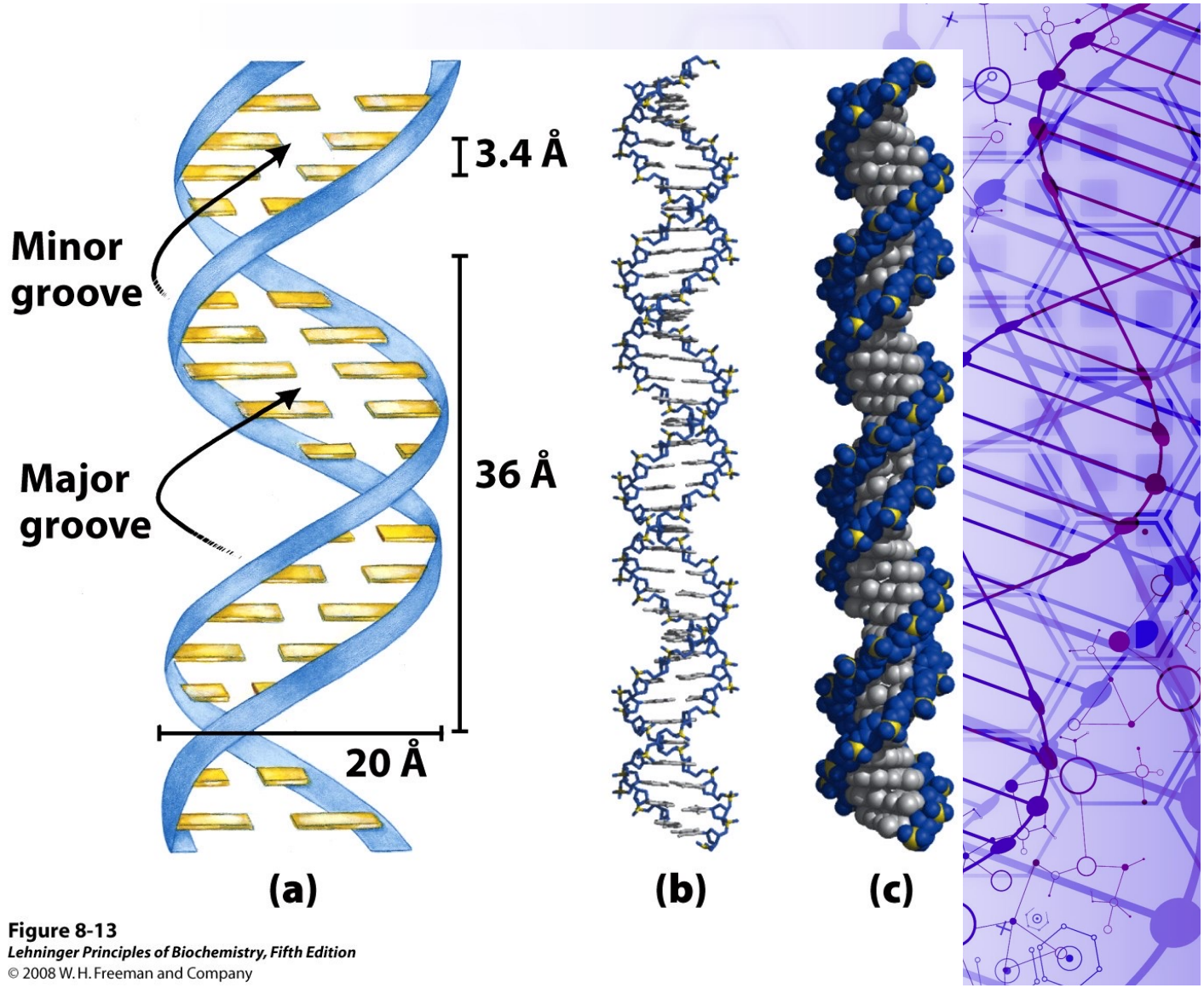
Duplex structures of DNA



B-form DNA

Figure 8-13
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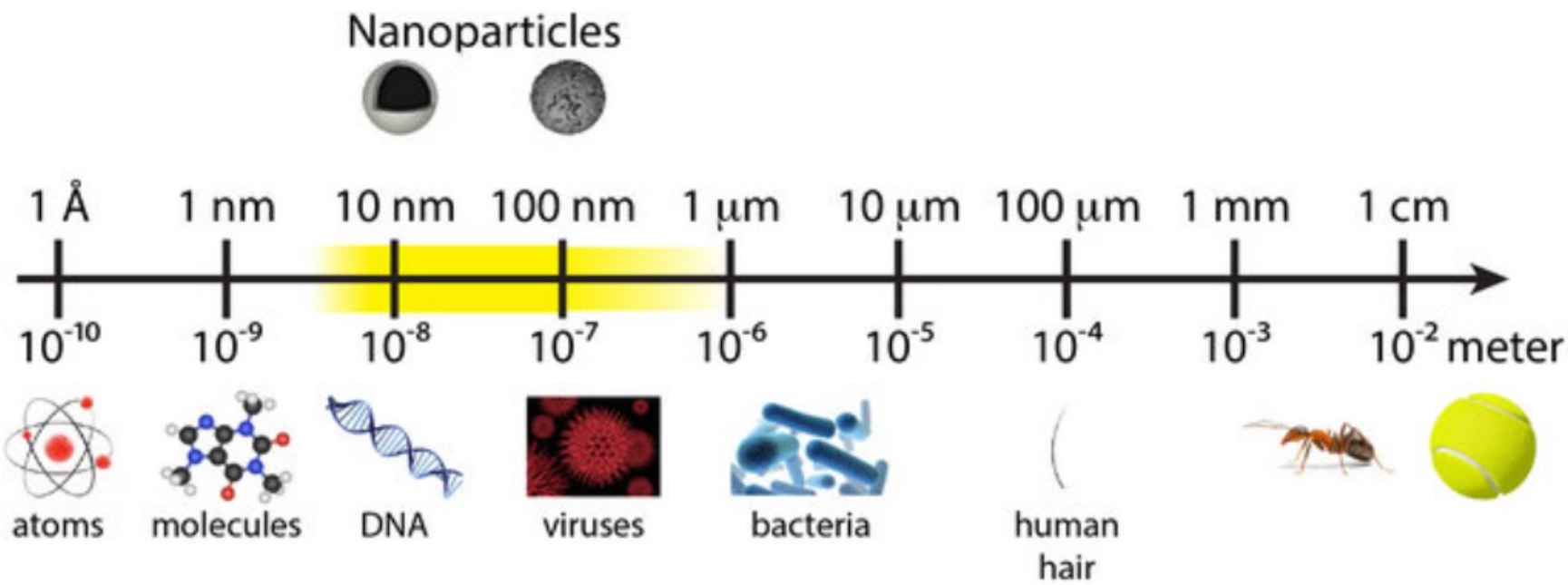
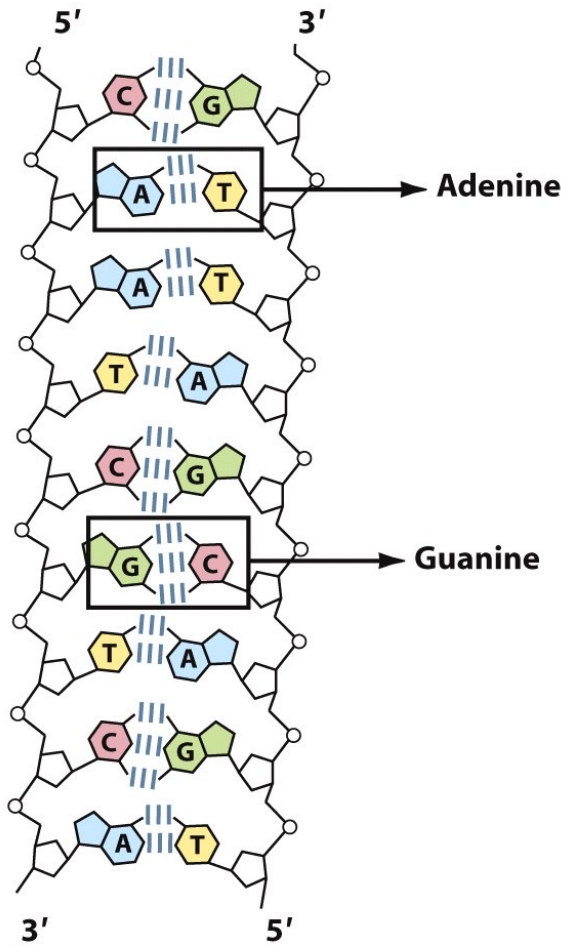
Length scale





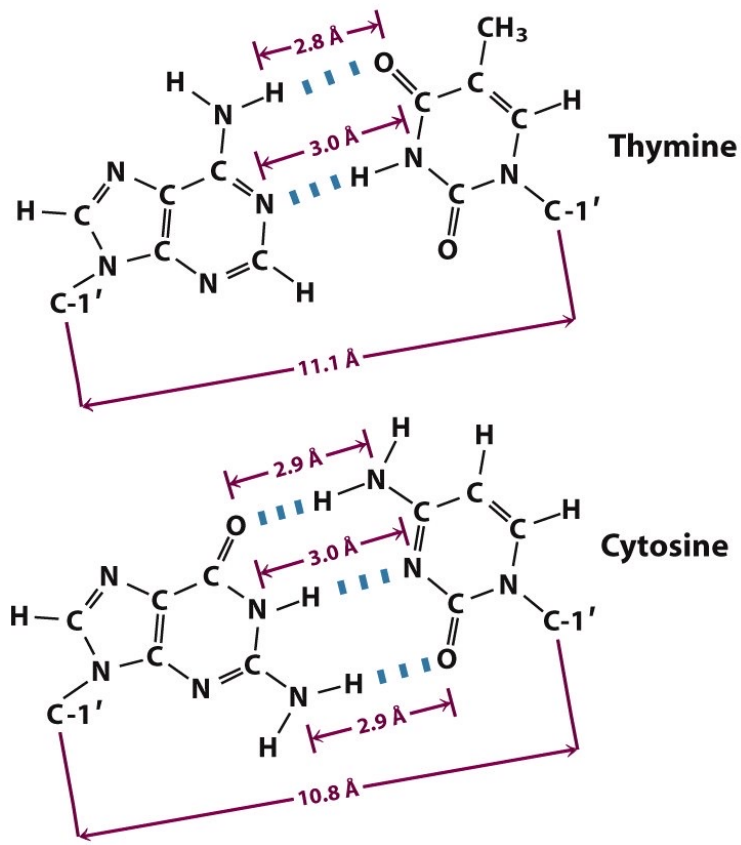


Figure 8-11
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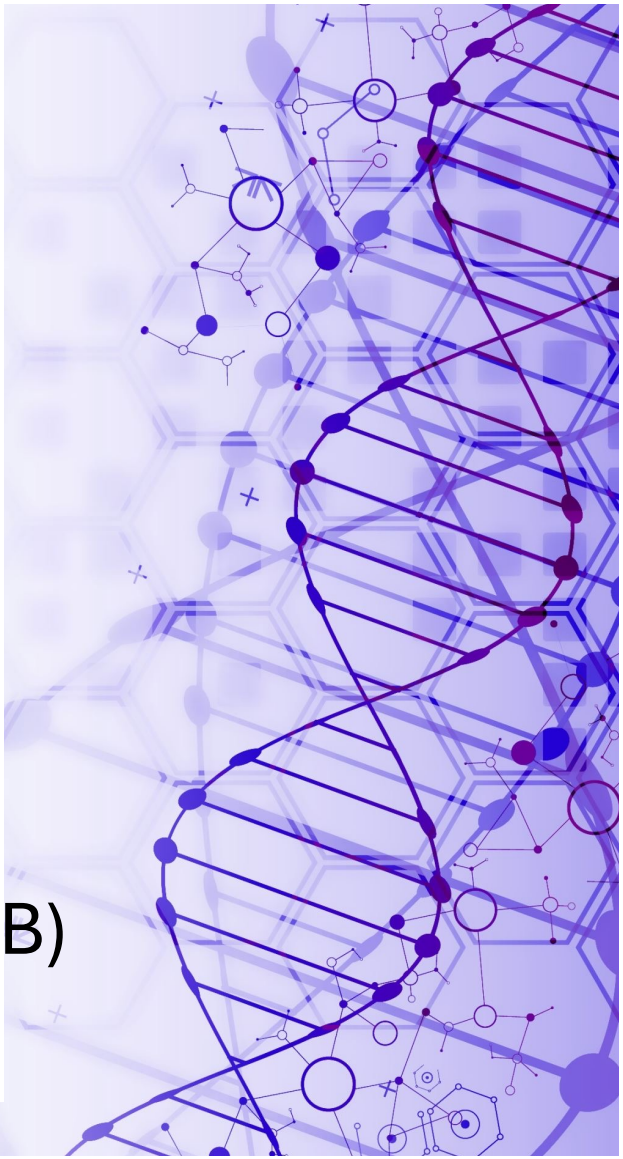
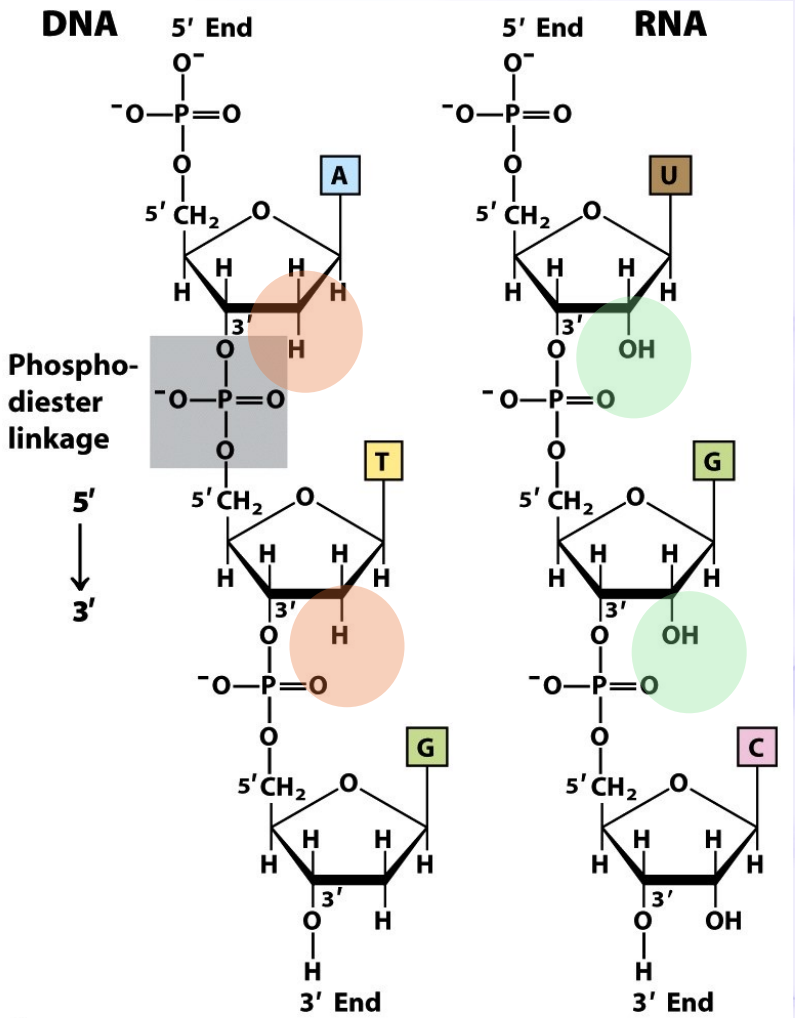
G ≡ C pair (three HB)

A = T pair (two HB)



Difference between DNA and RNA backbone

DNA bases: G C A T
RNA bases: G C A U



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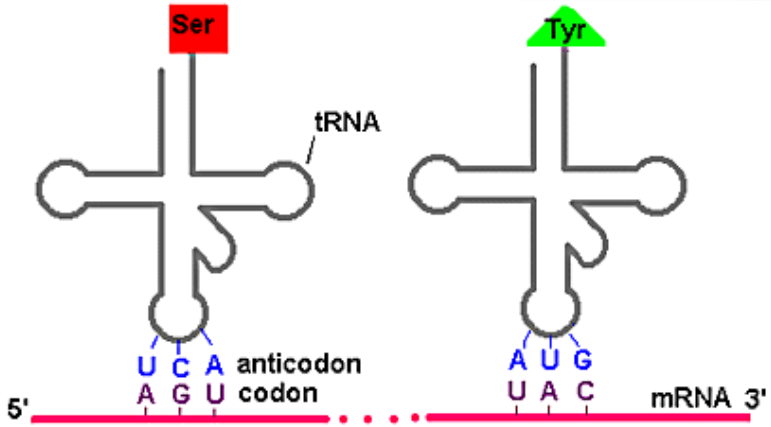


2nd base in codon

1st base in codon	U	C	A	G	3rd base in codon
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	STOP	STOP	A
	Leu	Ser	STOP	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

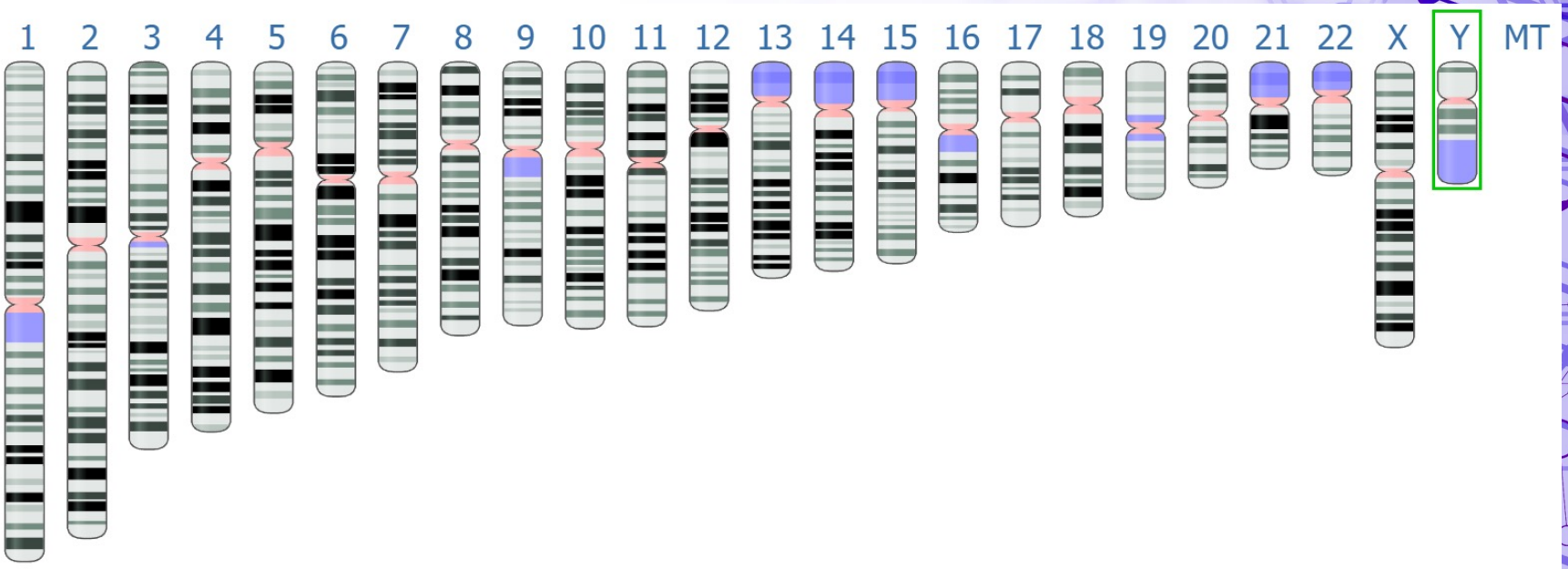
The Genetic Code

Cracked on 1966

- The genetic codon table is nearly universal, with some changes from species to species
- Mitochondria has a somewhat different codon table.

First Human Genome Project

- Started in 1990, completed by 2003;
- Initially planned for 20-year project, $3 billion USD effort



Link to Genome data viewer: https://www.ncbi.nlm.nih.gov/genome/gdv/browser/genome/?id=GCF_000001405.40

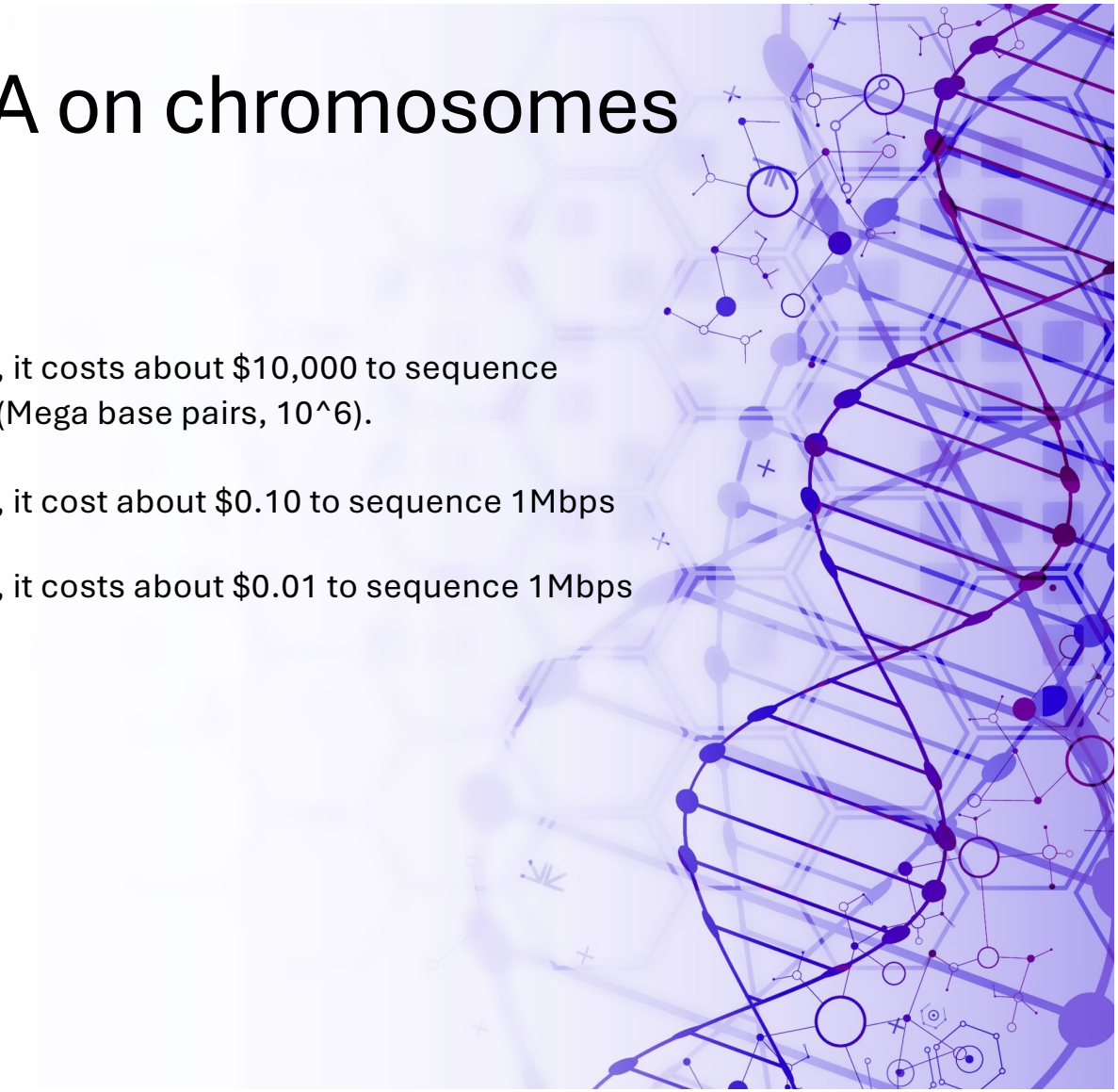
Sequencing all DNA on chromosomes

Chromoso	Base Pairs (bp)
1	248,956,422
2	242,193,529
3	198,295,559
4	190,214,555
5	181,538,259
6	170,805,979
7	159,345,973
8	145,138,636
9	138,394,717
10	133,797,422
11	135,086,622
12	133,275,309
13	114,364,328
14	107,043,718
15	101,991,189
16	90,338,345
17	83,257,441
18	80,373,285
19	58,617,616
20	64,444,167
21	46,709,983
22	50,818,468
X	156,040,895
Y	57,227,415

In 2001, it costs about $10,000 to sequence 1Mbps (Mega base pairs, 10^6).

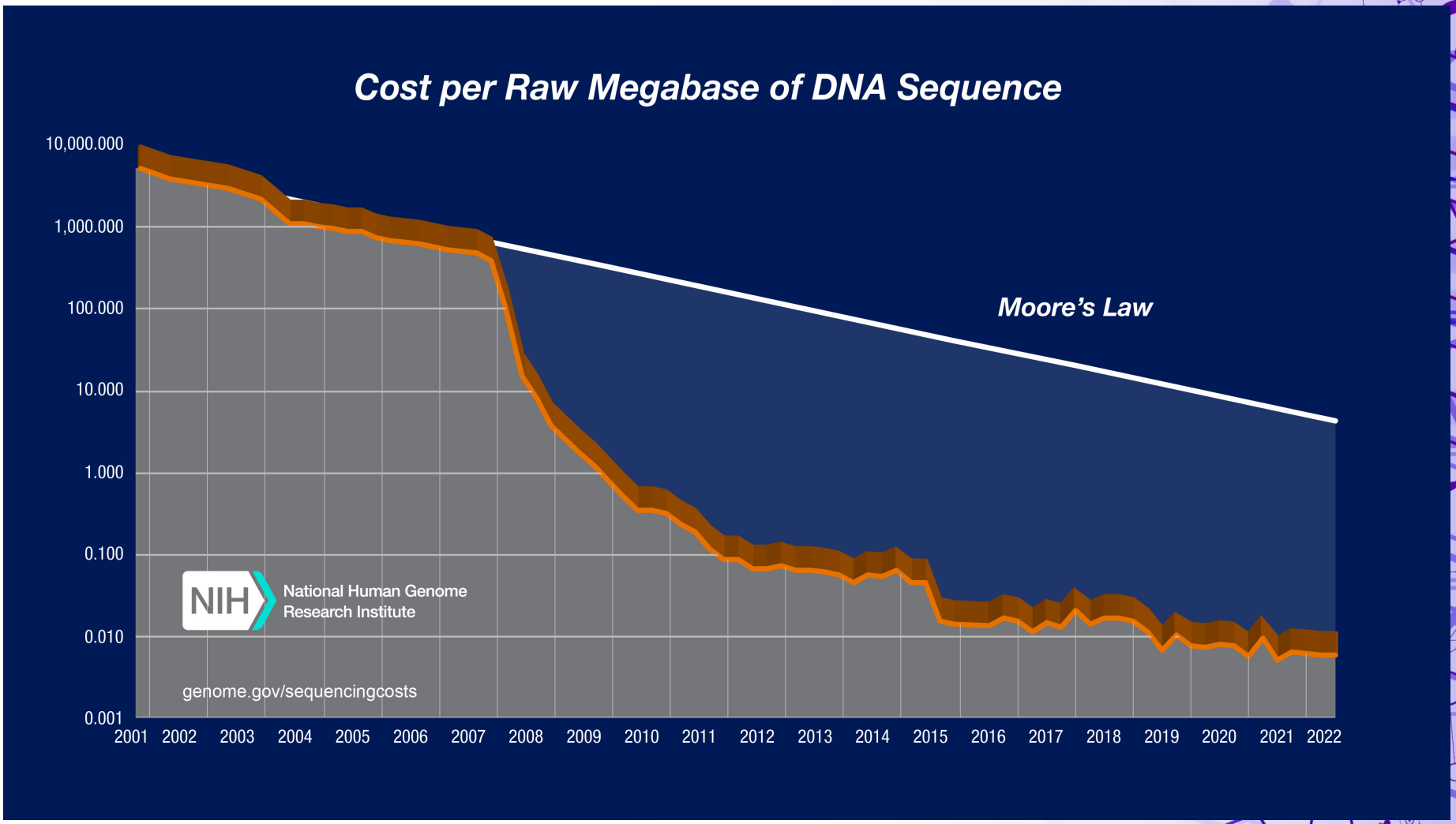
In 2011, it cost about $0.10 to sequence 1Mbps

In 2021, it costs about $0.01 to sequence 1Mbps



Activity: Jupyter Notebook

- Install and launch Jupyter notebook
- Use Jupyter notebook to plot some functions, y=x^2
- Use Jupyter notebook to plot some data contained in csv file
- Write a small code to calculate how long DNA on chromosome 1 if DNA is stretched out completely (each basepair length is 3.4An
- Write a small code to calculate the cost to sequence whole genome, with the cost of 1Mbps being varied from $10,000 to $1, to $0.10 and $0.01



https://www.genome.gov/about-genomics/fact-sheets/DNA-Sequencing-Costs-Data

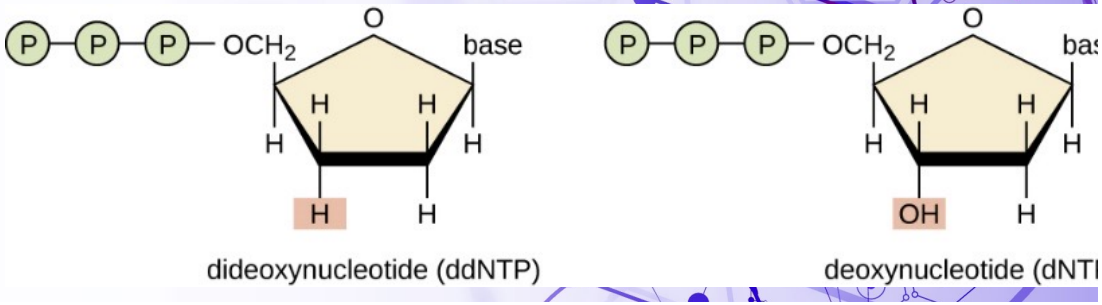
DNA sequencing Technologies

- Sanger Sequencing Method(kidding video) https://www.youtube.com/watch?v=dVRB4CaLizc

DNA polymerase: -- synthesizes new strands of DNA by adding nucleotides to a pre-existing strand (template) during **DNA replication**

ddNTP (dideoxynucleotide)-terminate chain growth

Fluorescent-labeled ddNTP

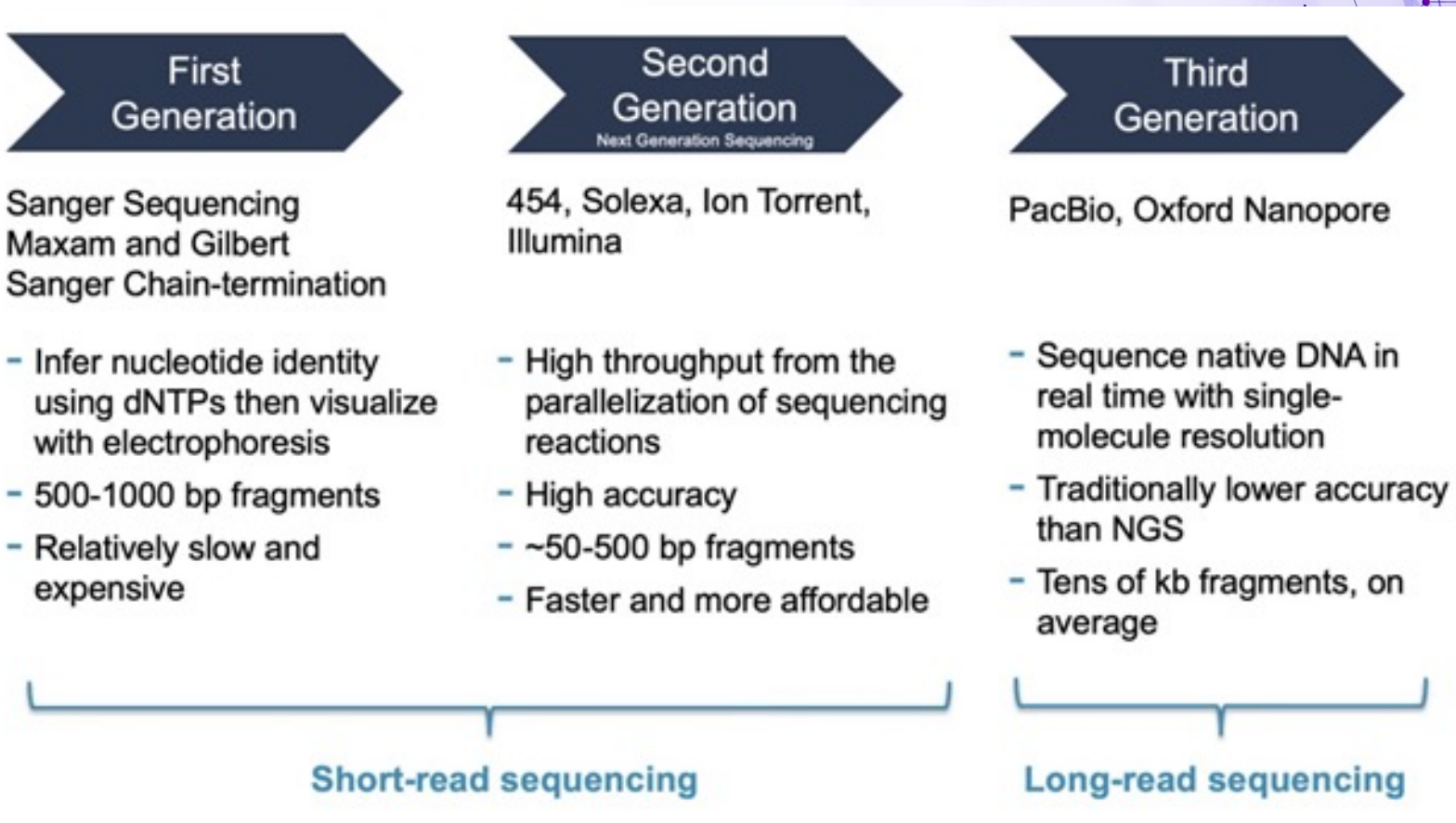


Sanger sequencing

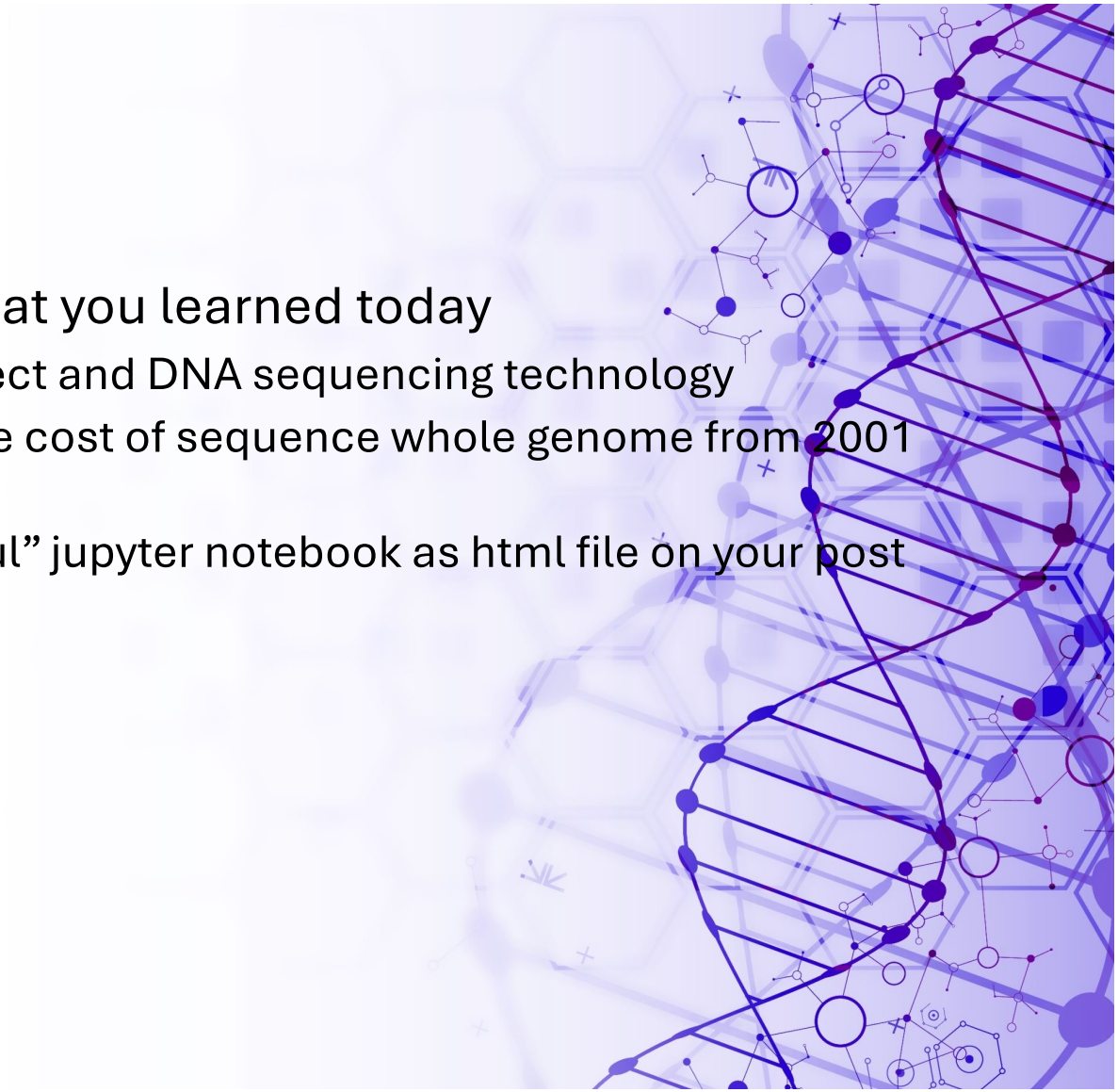
- Why it can only sequence up to 1000 bps?
- Because you need to line up DNA strand and resolve the length one single base pairs
- DNA strand is resolved through "**Gel electrophoresis" , which can only resolve well up to that length**

- **Work around: Shotgun sequencing:** Break large DNA into smaller fragments, sequence them individually, then assemble computationally.
- Sequence Assembly was the first primary computational task

DNA sequencing technology evolution



Ref: The Future of DNA Sequencing & How it impacts the Job Market!

Activity

- Write a little post about what you learned today
  - about Human genome project and DNA sequencing technology
  - Show your calculation of the cost of sequence whole genome from 2001 to 2021
  - Finally post your “successful” jupyter notebook as html file on your post as a link